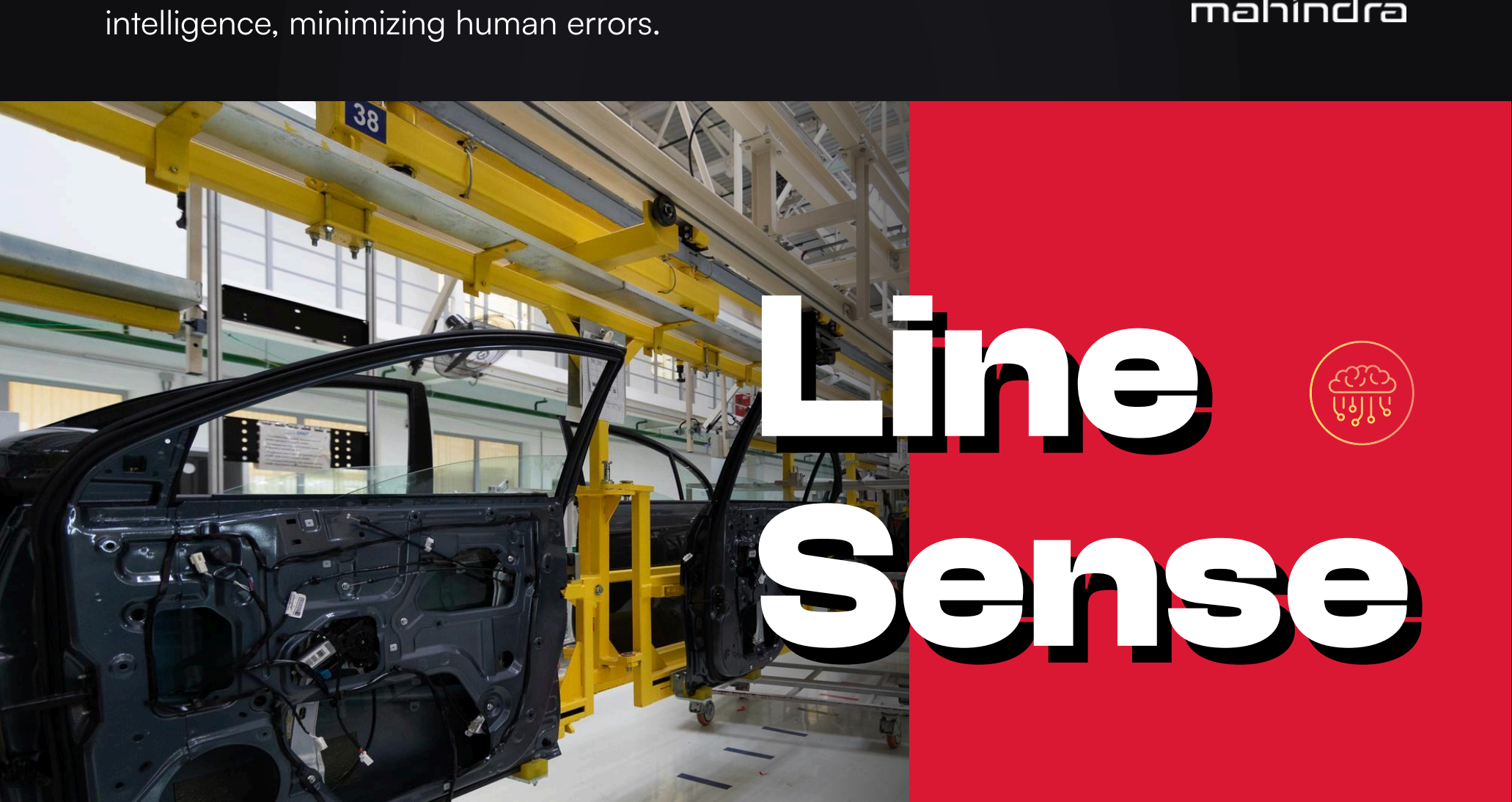
Mihir Vaidya

Hire Me

Powering assembly line efficiency with artificial intelligence, minimizing human errors.

TECH mahindra



OVERVIEW

In the realm of manufacturing excellence, precision and quality are paramount. Welcome to the world of "LineSense," where we've redefined the assembly line landscape.

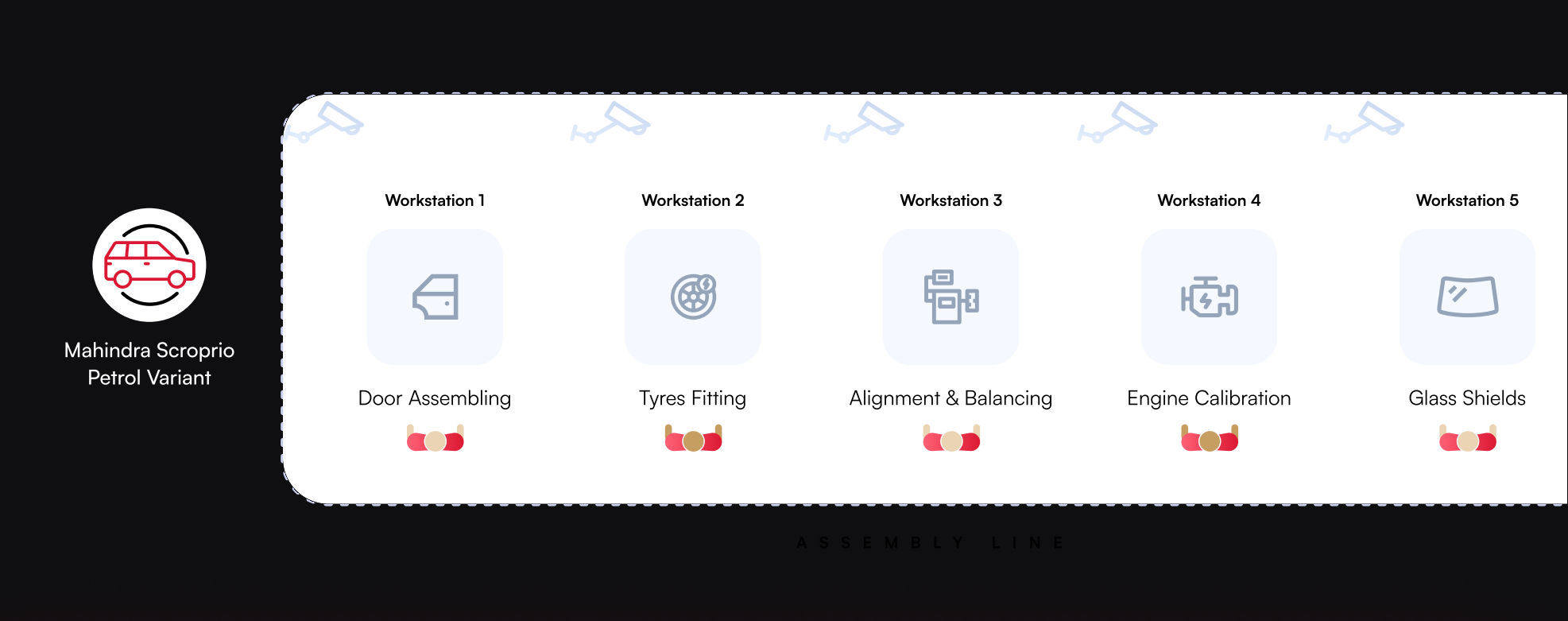
In this transformative project, we've harnessed the power of advanced technology, leveraging the combination of cameras and sensors to infuse artificial intelligence into most of the step of the manufacturing journey.

Smarter Assembly Lines

Minimised Errors

# Setup

For an instance, consider the product Mahindra Scorpio Petrol Variant


Mahindra Scorpio Petrol Variant
Workstation 1
Door Assembling
Workstation 2
Tyres Fitting
Workstation 3
Alignment & Balancing
Workstation 4
Engine Calibration
Workstation 5
Glass Shields

# Story

The journey unfolded for our heroes.

Data Scientist · Annotators · Validators

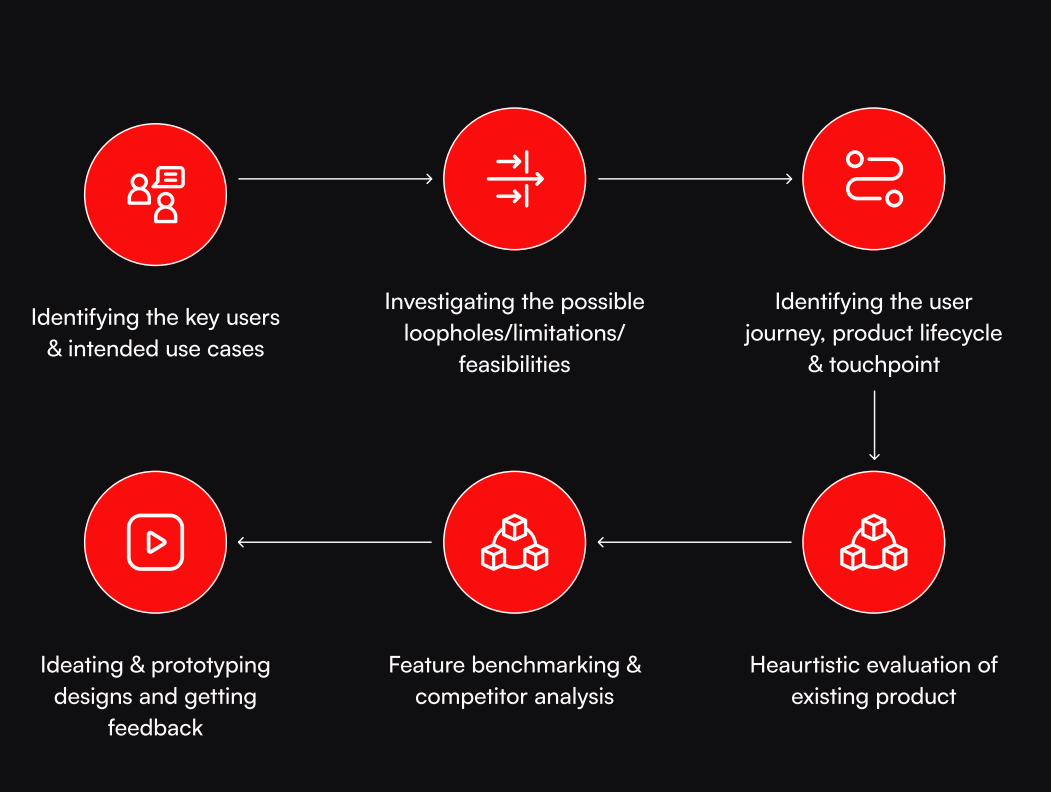
| | | | | |
|---|---|---|---|---|
| 1. Create Data set. | 2. Annotate agents and actions. | 3. Validate annotated data. | 4. Develop training model. | 5. Test & deploy model |
| Data Scientist | Annotators | Validators | Data Scientist | Data Scientist |

BUSINESS GOALS

The project aims to enhance manufacturing efficiency, improve product quality, reduce operational costs, and achieve scalability and adaptability in assembly line operations.

Effecient · Scalable · Adaptable

APPROACH



Identifying the key users & intended use cases

Investigating the possible loopholes/limitations/feasibilities

Identifying the user journey, product lifecycle & touchpoint

Ideating & prototyping designs and getting feedback

Feature benchmarking & competitor analysis

Heuristic evaluation of existing product

# Meet the heroes of our product.

**Maria Nicolas**

32, Female

Data Scientist

Alicia holds a master's degree in data science and has several years of experience in machine learning and computer vision. She is passionate about harnessing AI to solve real-world problems.

Challenges

Ensuring the quality and diversity of training data. Collaborating effectively with annotators and validators.

Motivation & Goals

Efficient dataset creation, Improving AI model accuracy, Successfully implementing AI solutions to enhance manufacturing.

**James Nortan**

28, Male

Annotator & Validator

James has a background in computer science and is skilled in image annotation. He has experience in accurately labeling data for AI model training.

Challenges

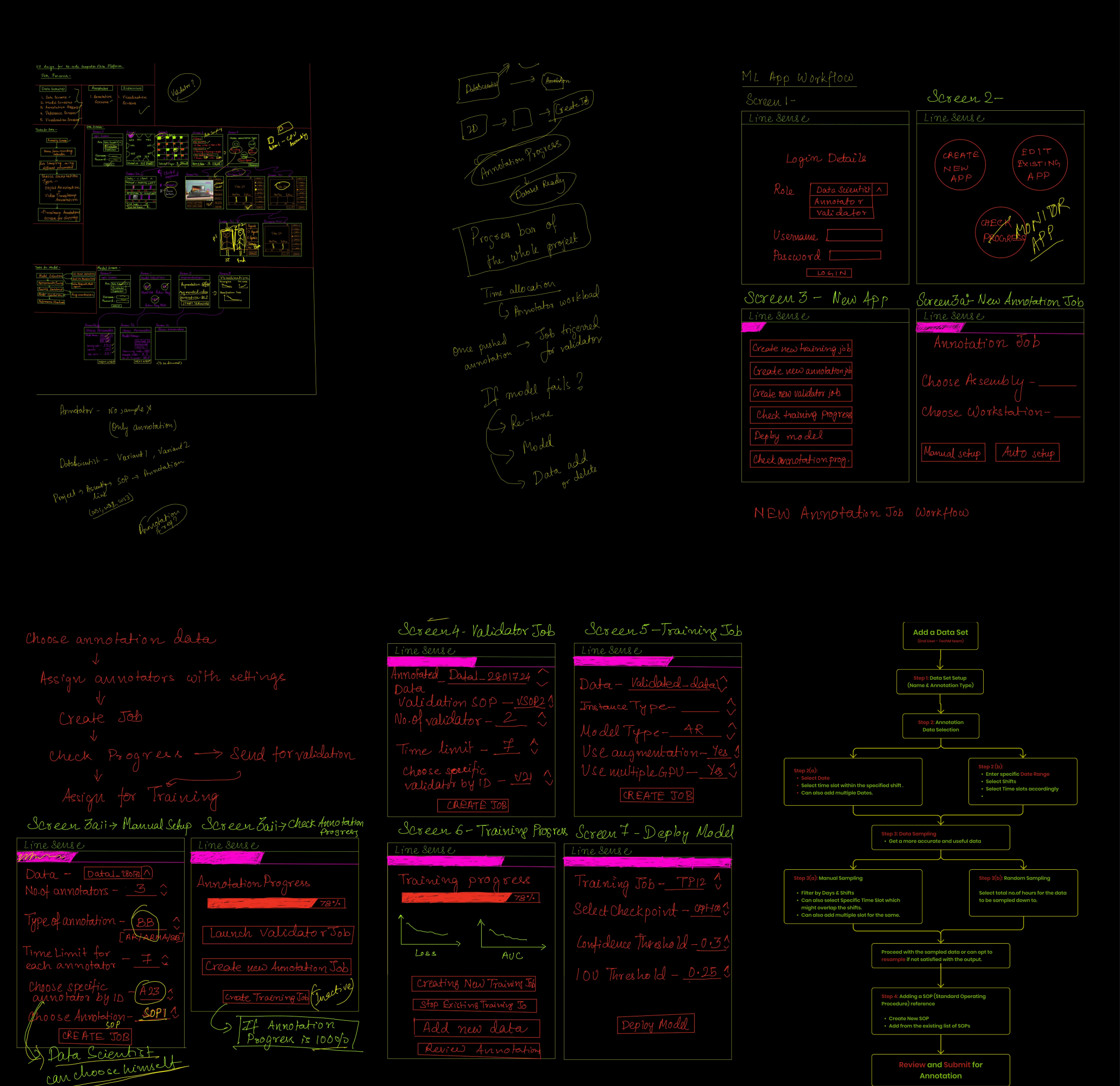
Maintaining annotation accuracy and consistency. Collaborating effectively within the team and the only source of truth to refer in case of complex labels/annotation.

Motivation & Goals

Providing high-quality annotations, Supporting data scientists in developing accurate AI models, Contributing to the improvement of manufacturing processes.

Synchronised operations for seamless progress.

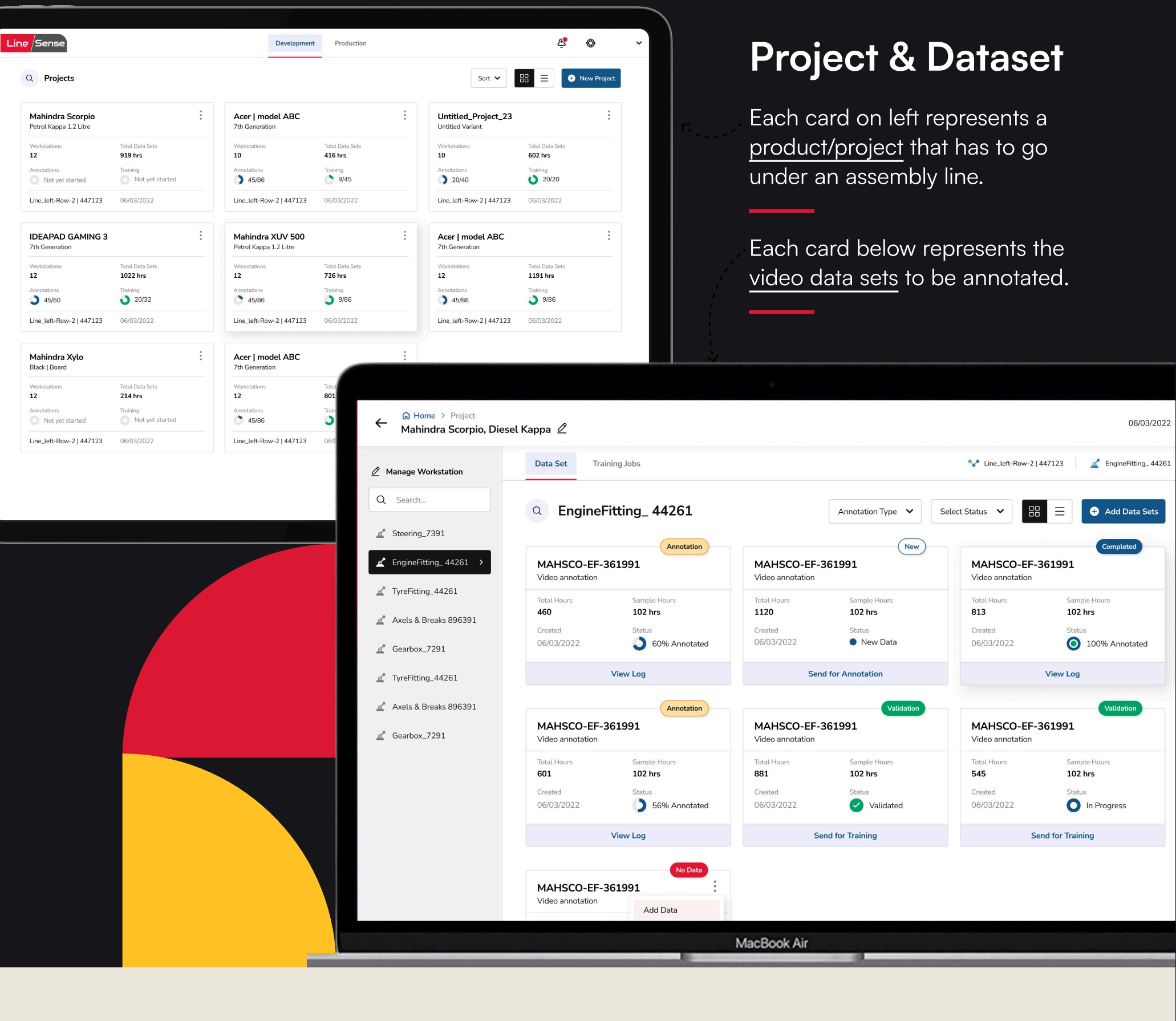
# workflow



## Project & Dataset

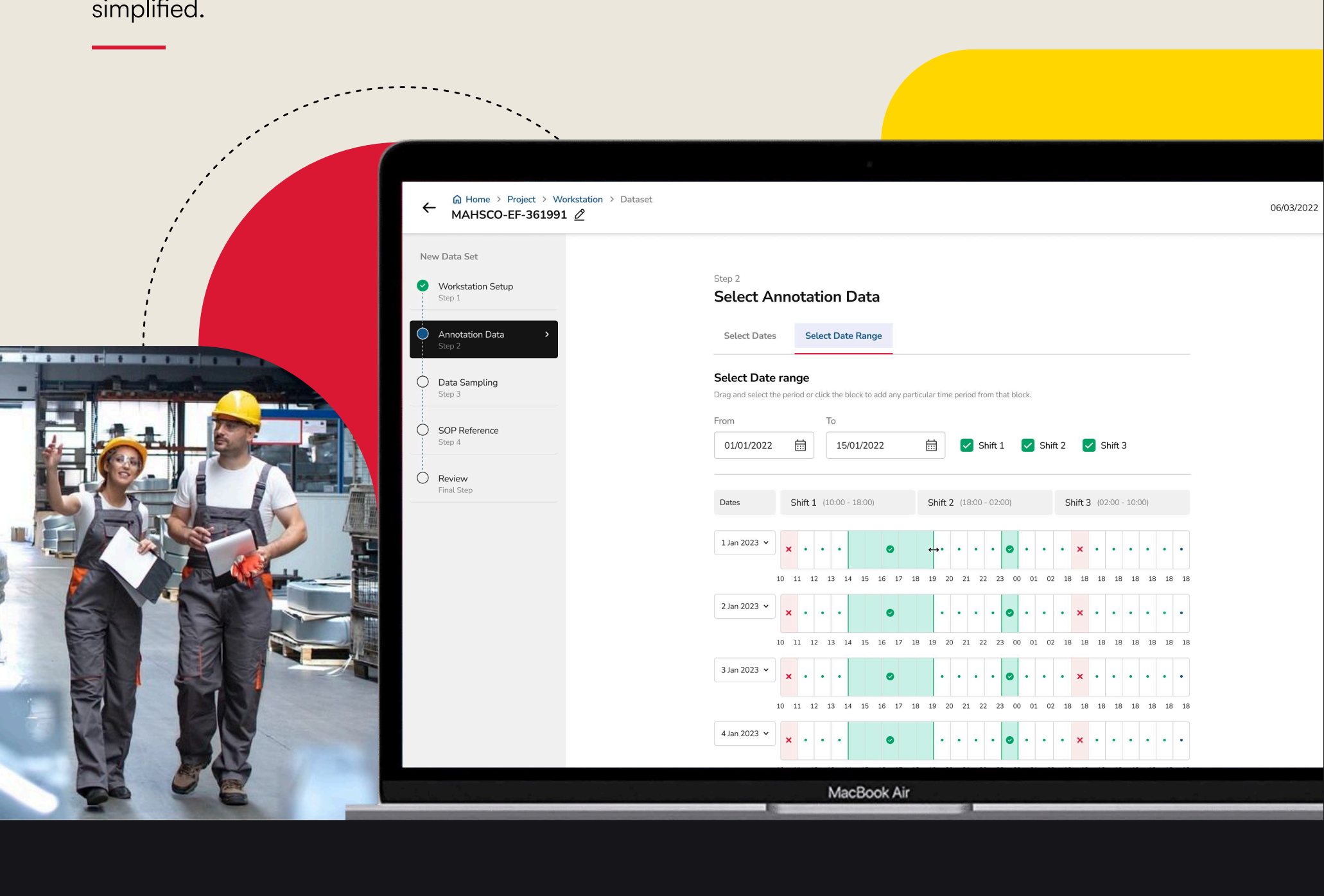
Each card on left represents a product/project that has to go under an assembly line.

Each card below represents the video data sets to be annotated.



## Data Sampling

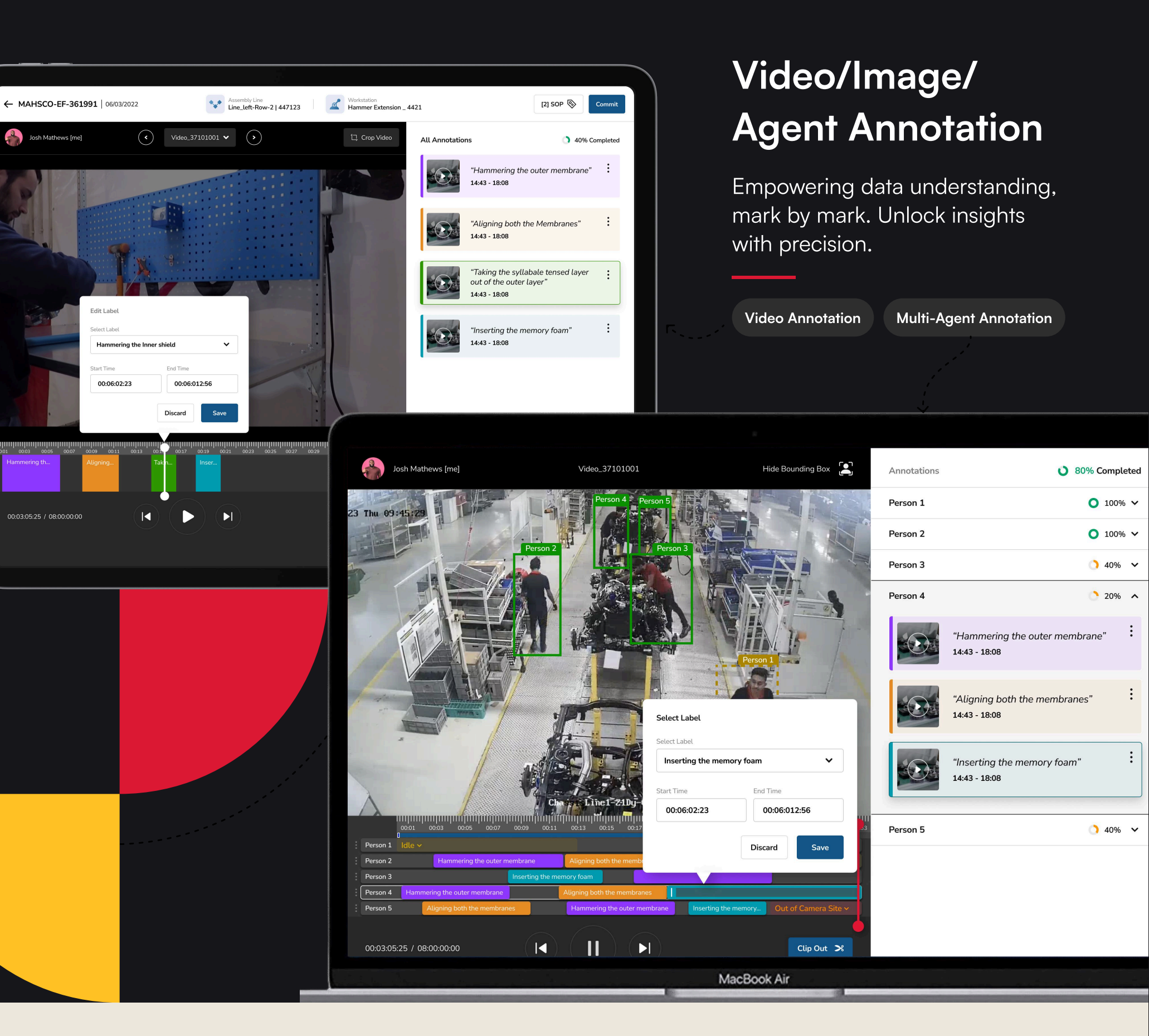
AI precision: Automated video sampling simplified.


Select Annotation Data
Select Date range

## Video/Image/Agent Annotation

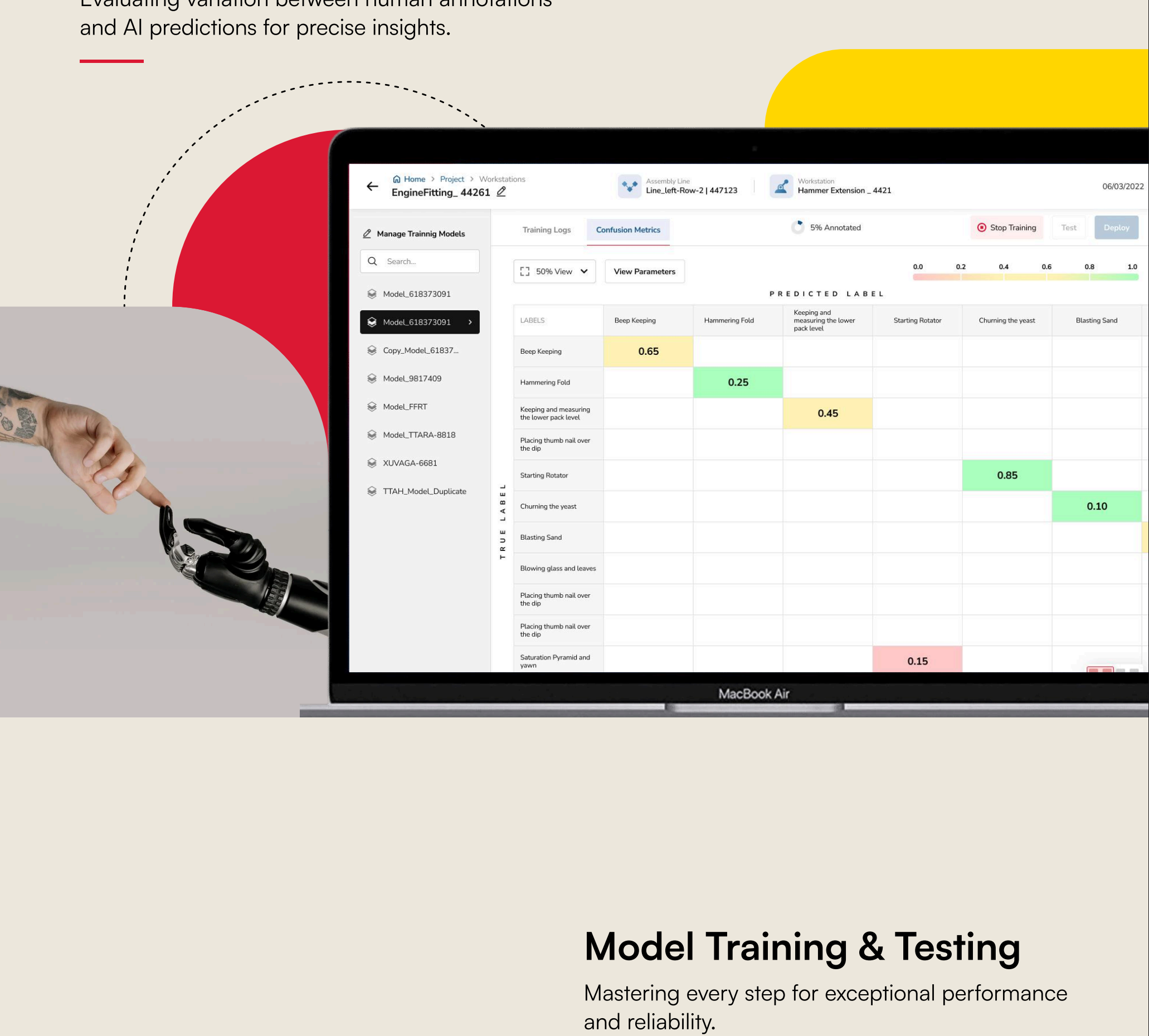
Empowering data understanding, mark by mark. Unlock insights with precision.

Video Annotation · Multi-Agent Annotation


"Hammering the outer membrane"
"Aligning both the membranes"
"Inserting the memory foam"
Select Label
Discard
Save

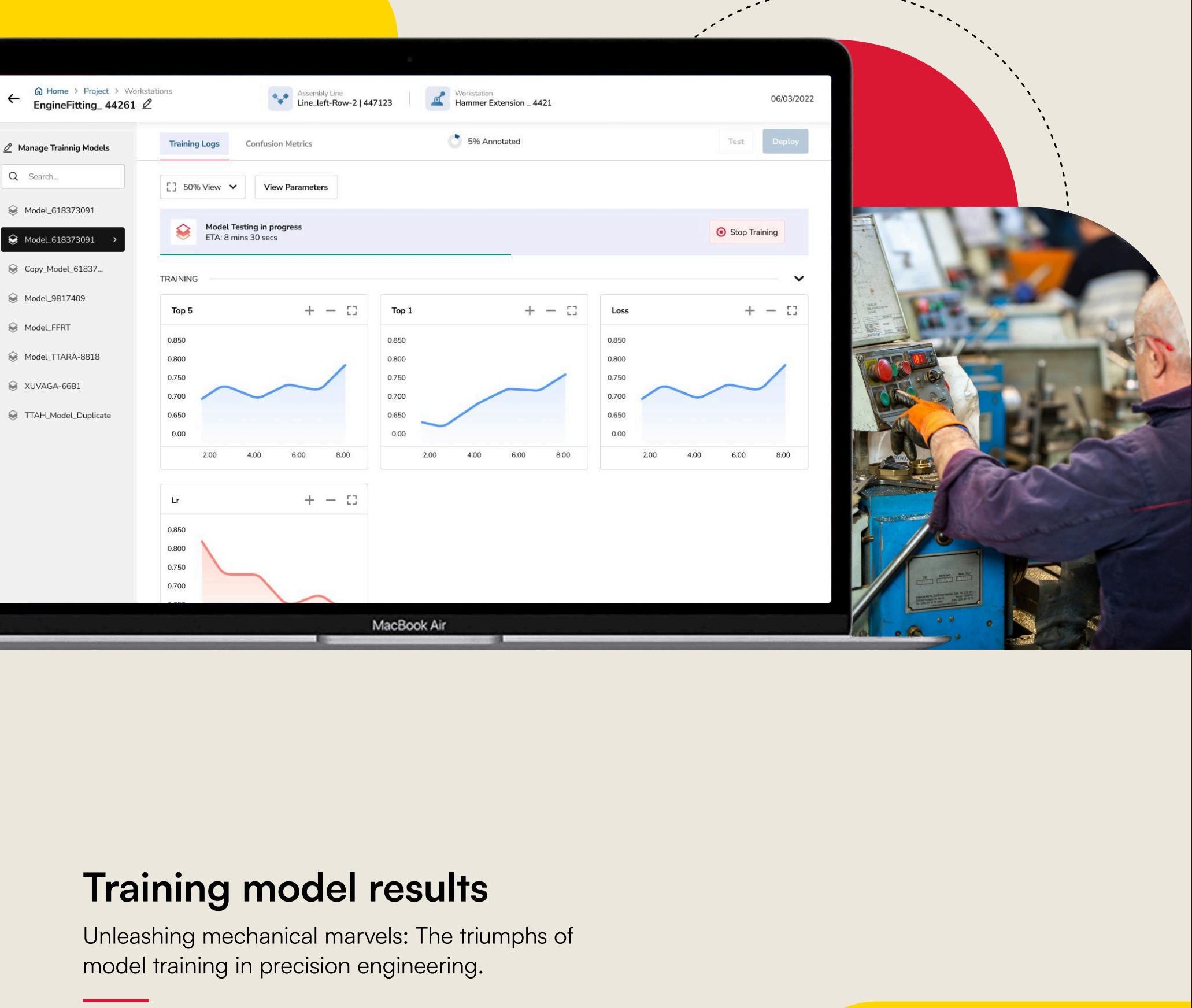
## AI Confusion Metrics

Evaluating variation between human annotations and AI predictions for precise insights.


PREDICTED LABEL
0.65
0.25
0.45
0.85
0.10
0.15
MacBook Air

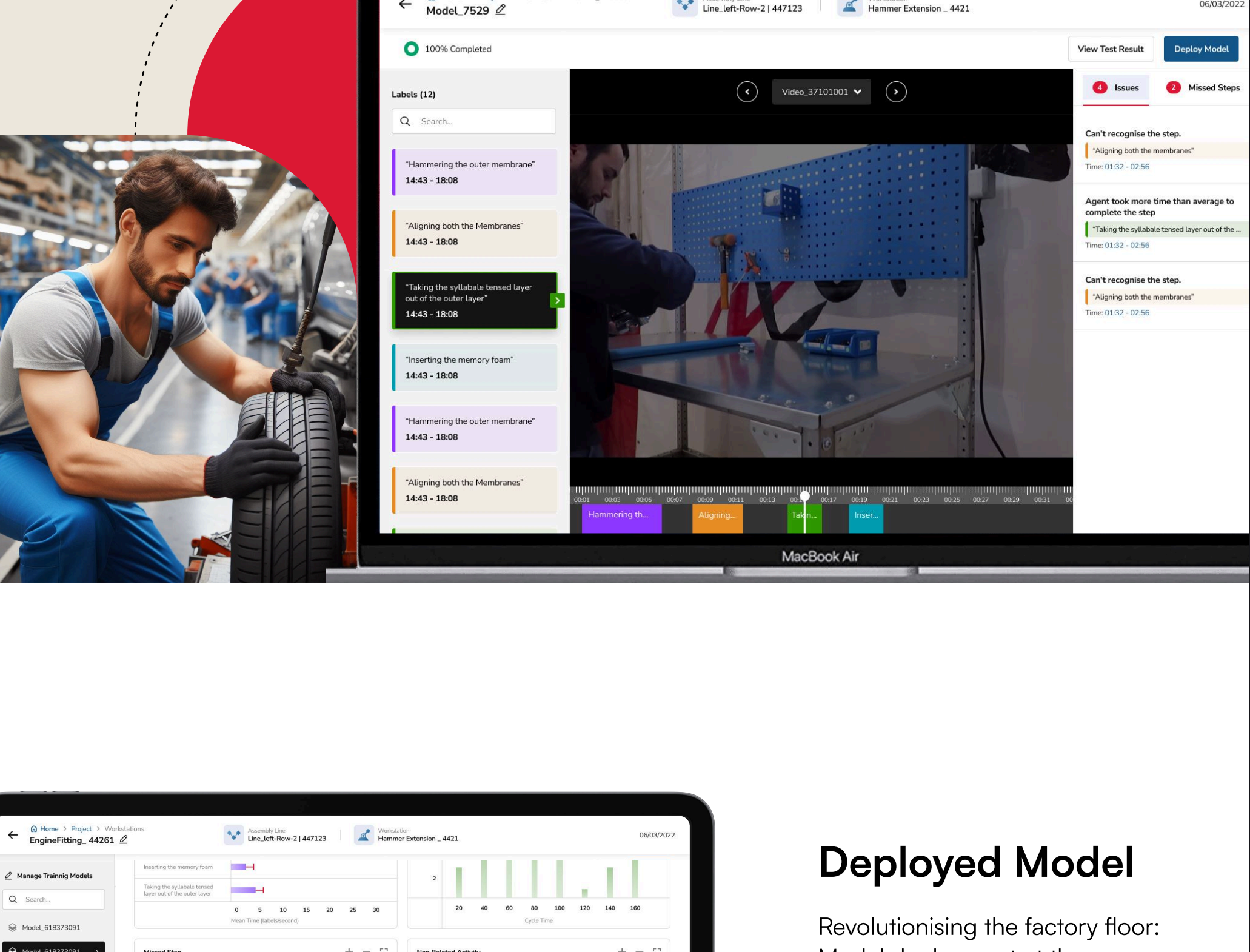
## Model Training & Testing

Mastering every step for exceptional performance and reliability.


Stop Training
Model Testing in progress
MacBook Air

## Training model results

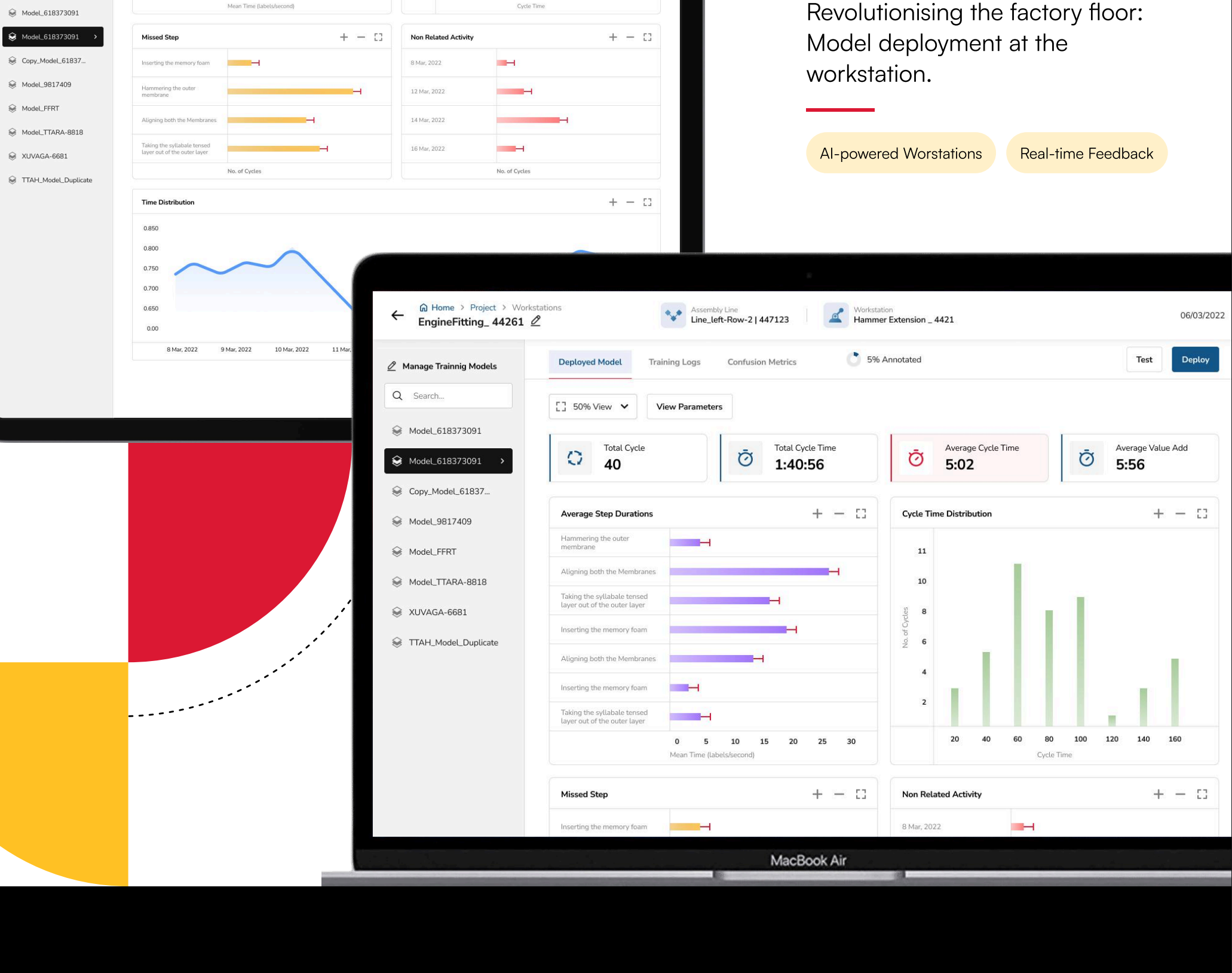
Unleashing mechanical marvels: The triumphs of model training in precision engineering.


100% Completion

## Deployed Model

Revolutionising the factory floor: Model deployment at the workstation.

AI-powered Workstations · Real-time Feedback


Deploy Model
Total Cycle 40
Total Cycle Time 1:40:56
Average Cycle Time 5:02
Average Value Add 5:56
MacBook Air

# Transforming Assembly Line Excellence

In the grand narrative of manufacturing, "LineSense" has emerged as the protagonist, reshaping the landscape of assembly line efficiency.

From the inception of data creation to the deployment of AI models on workstations, we've embarked on a journey that redefines precision in mechanical engineering.

DOMAIN LEARNINGS

With "LineSense," we've not only streamlined operations but also elevated product quality and reduced operational costs.

We stand at the crossroads of innovation, armed with scalability and adaptability, ready to write new chapters in manufacturing excellence.

The product is still under development and is going through a round of user testings. As they say...

Thank you for being a part of the "LineSense" story.

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